

Genuine question, how is AI justifiable in its water use?

Posted by u/someonetrying17 • 0 points • 6 comments

I'm just actually curious. How does the merit work out in terms of AI taking up lots of water resources, as well as mining resources for tech materials, vs the benefit of using AI and not hiring actual people? I'm not entirely against AI but I'm not super convinced either so I'm just looking for perspective

Comments

u/LengthyLegato114514 • 2 points • 2024-11-21 06:30 UTC

Because it doesn't need to.

"water" and "mineral" use/mining is tied to making chips, which is then tied to technological advancements

The exact same question could be asked if poisoning the earth with heavy metals and rare minerals is worth making chips for your phone, or computer, and it would be even more inane if you think about it.

"AI" is just big, big data sorted very well, liked with well designed neural networks, with computations that make processing data more efficient.

It's arguably just a **byproduct** of advances in data storages and neural networking, even.

The question then becomes "is the supposed water use \[which is never even shown to be significant\] worth the technological advancements in data storage, data modeling, data visualization and network infrastructure?"

Which is an **inane** question to ask when the world runs on technology.

u/Hugglebuns • 1 points • 2024-11-21 06:26 UTC

Well, its the same merit that goes into whether or not cobalt or titanium paint should be refined and produced. Traditional heavy metal pigments are known for their environmental toxicity and it isn't great. Especially when a lot of the colorful pigments like phtalos and quins are literally made from fossil fuel derivatives. Well, and acrylic base is literally just refined PLA, yes the elmers kind

u/webdev-dreamer • 3 points • 2024-11-21 06:19 UTC

It's justifiable in that it helps companies save a lot of \$\$\$ on labor costs, and it allows me to make cool AI art

u/someonetrying17 • 0 points • 2024-11-21 06:21 UTC

Well, I was talking about the environment more specifically and the way AI art takes up a lot of water resources that could potentially have been used for farming or consumption.

u/ninjasaid13 • 1 points • 2024-11-21 06:22 UTC

>the way AI art takes up a lot of water resources that could potentially have been used for farming or consumption.

How much do you think AI Art requires?

u/webdev-dreamer • 0 points • 2024-11-21 06:28 UTC

Ha! Got em! Just don't let these folks see articles like this one:

<https://www.forbes.com/sites/cindygordon/2024/02/25/ai-is-accelerating-the-loss-of-our-scarcest-natural-resource-water/>

u/searcher1k • 1 points • 2024-11-21 06:42 UTC

This article falls short due to glaring journalistic oversights, such as failing to properly explain scarcity, conflating unrelated issues, and avoiding direct, "apples-to-apples" comparisons. Notably, there's little to no discussion of other water-intensive industries, which could provide important context.

Here's a straightforward comparison: Earth holds 1.386 billion km³ of water, of which only 0.03% is liquid freshwater. That equates to approximately 4×10^{14} m³ of available freshwater. By 2027, AI water usage is projected at 6.6 billion m³, which represents just 0.001% of the total available liquid freshwater—a negligible fraction.

Now, consider truly water-intensive industries. Livestock farming, for instance, uses about 4,387 km³ (4,387 billion m³) annually—roughly 700 times more than AI is expected to consume in 2027.

The author, a corporate venture capitalist with an AI sales consulting background, seems to be fabricating a "problem" as a convenient talking point to promote her company's services.

u/someonetrying17 • 1 points • 2024-11-21 06:25 UTC

I've been reading more on this sub. It's not as much as I thought. Like I said I'm just looking for perspective not trying to snark or anything